

GRIDPULSE AI

Power Outage & Grid Equipment Failure Advisor

Detect. Explain. Recommend. Simulate. Act.

TEAM

Eat-Code-Sleep

TRACK

AI

TEAM MEMBERS

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THE PROBLEM

Outages are often predictable
but action comes too late

Early warning signals exist
but go unseen

Data is siloed across systems
and conditions

Maintenance is reactive or calendar-driven

Failures can cascade into
widespread outages

Operators lack a single,
actionable risk view



Power failures are often predictable — but acted on too late.

Utilities have data, but the signals live in separate systems.

FRAGMENTED SIGNALS

Temperature, vibration, DGA and partial-discharge data are rarely fused with weather.

REACTIVE MAINTENANCE

Calendar-based maintenance can miss degradation signatures that appear weeks earlier.

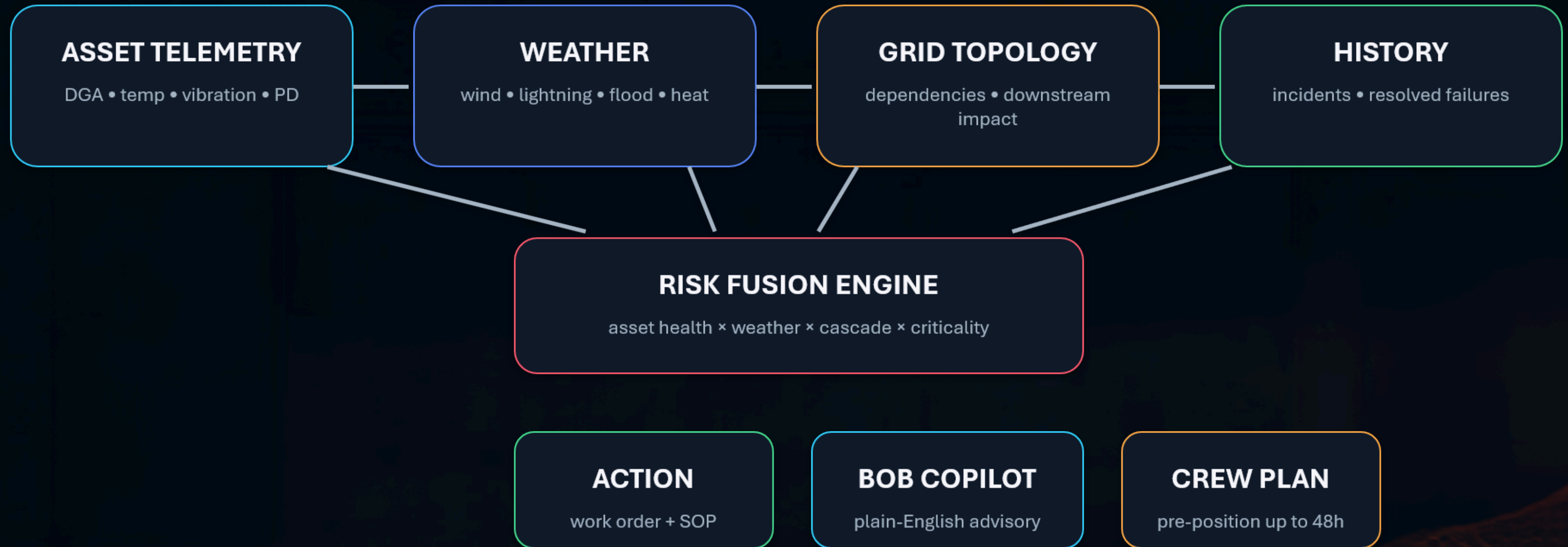
CASCADE IMPACT

One asset failure can propagate through the grid and disrupt critical facilities.

The missing layer: a single risk view that tells operators **WHICH** asset is at risk, **WHY**, **WHEN**, and **WHAT TO DO**.

One operational copilot for predictive grid resilience.

GridPulse fuses four evidence layers into a ranked, explainable action plan.



ARCHITECTURE

A modular AI platform that
turns data into outage foresight

React Dashboard
real-time visibility

FastAPI
high-performance backend

PostgreSQL
reliable relational data

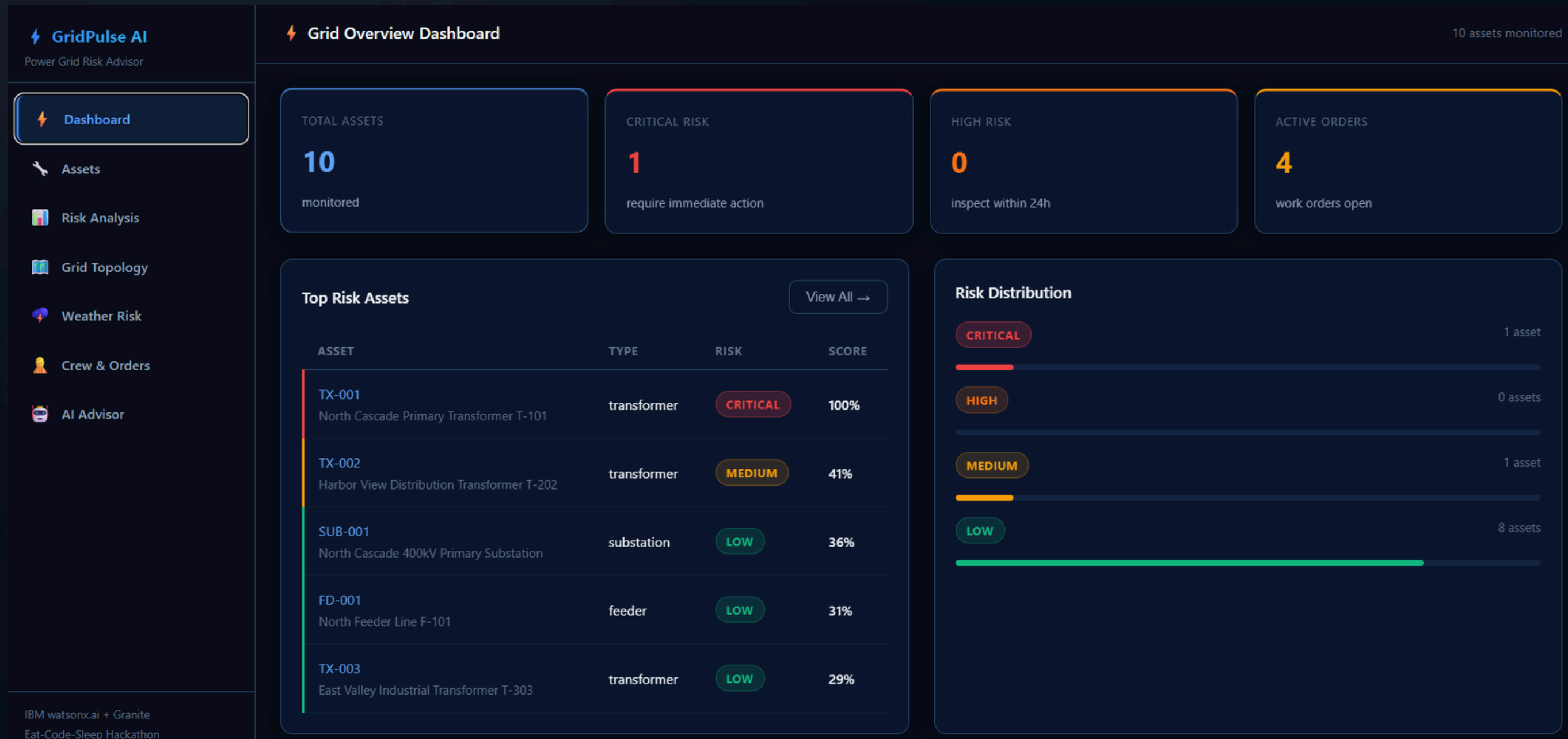
Neo4j
connected grid intelligence

IBM watsonx.ai / Granite
advanced AI foundation

Risk engines
DGA analysis
temperature/vibration analysis
weather risk
cascade/topology analysis



DEMO / KEY FEATURE



See overall grid health at a glance

Identify the assets that matter most

Understand the real drivers of risk

Get clear AI reasoning you can trust

Act with recommended steps and priority

DETECT → EXPLAIN → PRIORITIZE → DISPATCH

IBM TECHNOLOGIES

Powering GridPulse AI end to end

IBM Bob

AI-assisted development that
accelerates and elevates delivery

IBM watsonx.ai

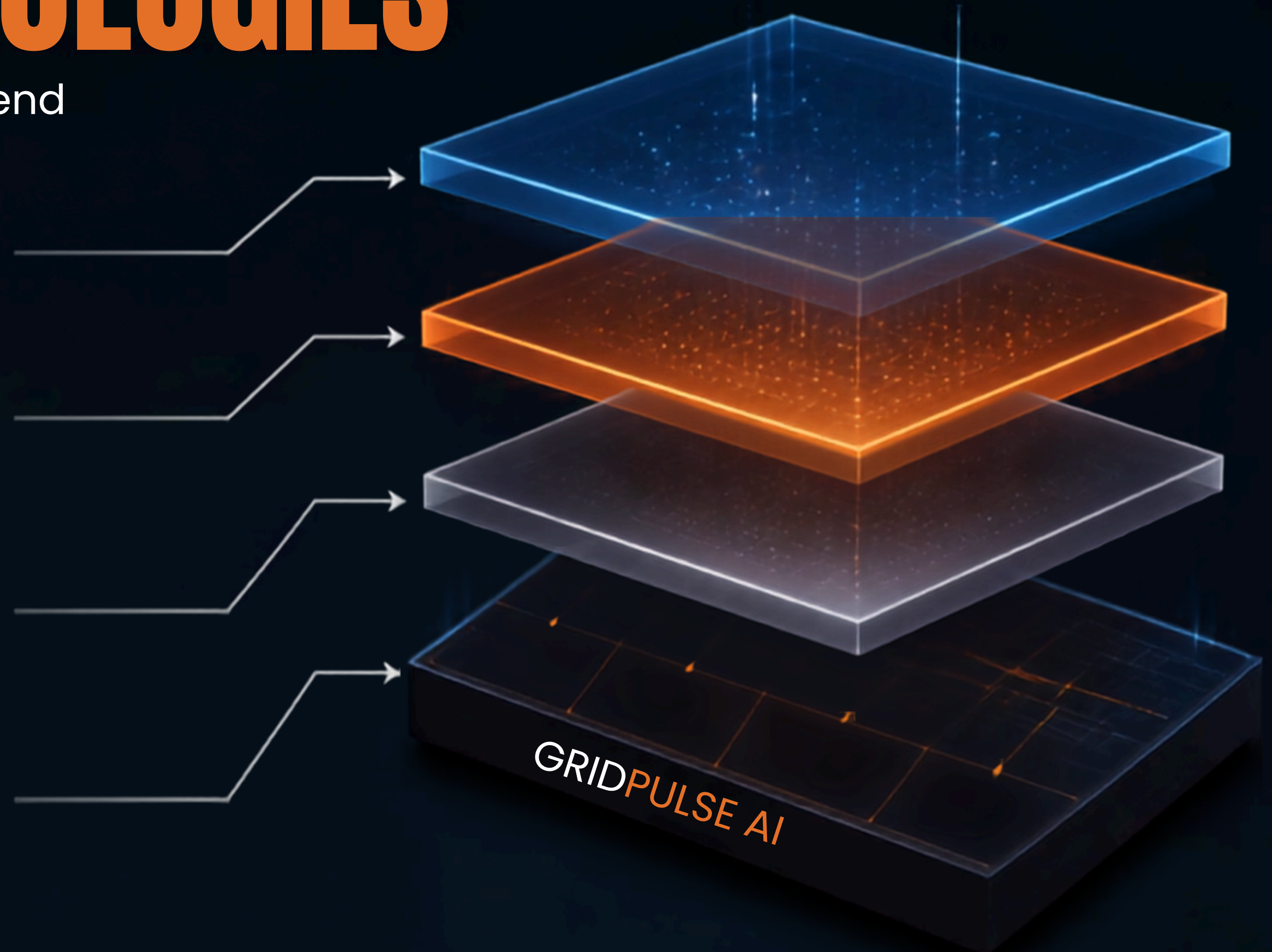
AI reasoning and advisory layer that
turns data into insight

IBM Granite

LLM-powered recommendations
grounded in grid knowledge

Application workflow

Integrated into risk detection, explanation,
and decision workflows



RESULTS & IMPACT

95

automated
tests passed

48

hour crew
pre-positioning
concept

4

evidence
layers
combined

1

unified
operator
decision view

expected impact

earlier
failure
detection



better
maintenance
prioritization



faster
field
response



reduced
avoidable
outage risk



THE TEAM

Darshan Thummar

Team Lead, Full Stack & AI Architecture

Shreeja Upadhyay

Backend & Risk Engine

Kishan Vadsola

Frontend & Dashboard

Vishv Undavia

AI Integration, Testing & Demo

Detect. Explain. Prioritize. Act.

